

The K-Nearest Neighbors (KNN) algorithm is one of the simplest and most intuitive machine learning algorithms used for classification and regression tasks. Below is a step-by-step explanation of how the KNN algorithm works for classification.

## Step 1: Load the Data

Before using the KNN algorithm, you need to have a dataset that consists of labeled data. Each data point in the dataset should have a feature vector and a corresponding class label.

## Step 2: Choose the Value of K

The "K" in KNN refers to the number of nearest neighbors that the algorithm will consider when making a classification decision. You must choose a value for K, which is typically an odd number to avoid ties in binary classification.

- **Small K:** More sensitive to noise but may capture complex patterns.
- **Large K:** More stable but might miss smaller patterns.

## Step 3: Calculate the Distance

For each data point in the test set (the data you want to classify), calculate the distance between this point and all points in the training set. The most common distance metric used is the Euclidean distance, but other metrics like Manhattan or Minkowski distances can also be used.

**Euclidean Distance Formula:**

$$d(p, q) = \sqrt{\sum_{i=1}^n (p_i - q_i)^2}$$

Where:

- $p_i$  and  $q_i$  are the feature values of the two points.
- $n$  is the number of features.

## Step 4: Identify the Nearest Neighbors

Once the distances are calculated, sort the training data based on the distance from the test point. The K closest data points are selected as the nearest neighbors.

## Step 5: Assign the Class Label

For classification, KNN uses a majority vote mechanism among the K nearest neighbors. The class label that appears most frequently among these neighbors is assigned to the test point.

- **Example:** If you have K=5, and 3 neighbors belong to class A while 2 neighbors belong to class B, the test point will be classified as class A.

## Step 6: Evaluate the Model

After assigning class labels to all test points, you can evaluate the performance of your KNN classifier using metrics such as accuracy, precision, recall, and F1-score. Cross-validation is often used to determine the best value of K.

## Step 7: (Optional) Weighted Voting

In some cases, you might want to give more weight to closer neighbors. This is done by using a weighted voting mechanism, where the contribution of each neighbor is inversely proportional to its distance from the test point.

**Weighted Voting Formula:**

$$\text{weight} = \frac{1}{d(p,q)}$$

## Step 8: Adjust and Repeat

If the model's performance is not satisfactory, you may need to adjust the value of K or use a different distance metric. Re-train the model and evaluate it again.

Let's work through a simple example using the KNN algorithm for classification with a small dataset.

## Example Dataset

Consider the following dataset with two features (**x1**, **x2**) and a class label (**Class**):

| Data Point | X1 | X2 | Class |
|------------|----|----|-------|
| A          | 1  | 2  | 0     |
| B          | 2  | 3  | 0     |
| C          | 3  | 1  | 1     |
| D          | 6  | 5  | 1     |
| E          | 7  | 8  | 1     |

### Step 1: Define the Test Point

Let's say we want to classify a new point **F** with coordinates **x1** = 3 and **x2** = 4.

### Step 2: Choose the Value of K

Assume we choose  $K = 3$ .

### Step 3: Calculate the Distance

We'll calculate the Euclidean distance between point **F** and all points in the dataset.

**Euclidean Distance Formula:**

$$d(p, q) = \sqrt{(X1_p - X1_q)^2 + (X2_p - X2_q)^2}$$

**Distance between F and A:**

$$d(F, A) = \sqrt{(3 - 1)^2 + (4 - 2)^2} \quad d(F, A) = \sqrt{(2)^2 + (2)^2} \quad d(F, A) = \sqrt{4 + 4} \quad d(F, A) = \sqrt{8} \approx 2.83$$

**Distance between F and B:**

$$d(F, B) = \sqrt{(3 - 2)^2 + (4 - 3)^2} \quad d(F, B) = \sqrt{(1)^2 + (1)^2} \quad d(F, B) = \sqrt{1 + 1} \quad d(F, B) = \sqrt{2} \approx 1.41$$

**Distance between F and C:**

$$d(F, C) = \sqrt{(3 - 3)^2 + (4 - 1)^2} \quad d(F, C) = \sqrt{(0)^2 + (3)^2} \quad d(F, C) = \sqrt{0 + 9} \quad d(F, C) = \sqrt{9} = 3.00$$

**Distance between F and D:**

$$d(F, D) = \sqrt{(3 - 6)^2 + (4 - 5)^2} \quad d(F, D) = \sqrt{(-3)^2 + (-1)^2} \quad d(F, D) = \sqrt{9 + 1} \\ d(F, D) = \sqrt{10} \approx 3.16$$

**Distance between F and E:**

$d(F, E) = \sqrt{(3 - 7)^2 + (4 - 8)^2}$     $d(F, E) = \sqrt{(-4)^2 + (-4)^2}$     $d(F, E) = \sqrt{16 + 16}$   
 $d(F, E) = \sqrt{32} \approx 5.66$

Step 4: Identify the Nearest Neighbors

Now, let's sort the distances:

| Data Point | Distance |
|------------|----------|
| B          | 1.41     |
| A          | 2.83     |
| C          | 3.00     |
| D          | 3.16     |
| E          | 5.66     |

The 3 nearest neighbors to point F are B, A, and C.

Step 5: Assign the Class Label

Let's now check the classes of the nearest neighbors:

- B has class 0.
- A has class 0.
- C has class 1.

The classes of the neighbors are {0, 0, 1}. Since the majority class is 0 (it appears twice), we assign class 0 to the test point F.

Final Result:

The KNN algorithm classifies the point F as belonging to class 0.

# K-Nearest Neighbors (KNN) Algorithm for Regression

KNN can also be used for regression tasks, where the goal is to predict a continuous output (a number) rather than a class label. The steps are similar to the classification process but with a different final step. Let's walk through the algorithm and an example mathematically.

## Step-by-Step KNN Algorithm for Regression

- 1. Load the Data:** Start with a dataset where each data point has features and a corresponding target value (instead of a class label).
- 2. Choose the Value of K:** Select the number of nearest neighbors,  $k$ .
- 3. Calculate the Distance:** Compute the distance between the new data point and all the points in the training dataset.
- 4. Identify the Nearest Neighbors:** Sort the distances and select the  $K$  nearest data points.
- 5. Predict the Output:** For regression, instead of voting, the algorithm predicts the output as the average (or weighted average) of the target values of the  $K$  nearest neighbors.
- 6. Evaluate the Model:** Use metrics such as Mean Squared Error (MSE) or Mean Absolute Error (MAE) to evaluate the model's performance.

## Mathematical Example of KNN for Regression

### Example Dataset

Consider the following dataset with two features ( $x_1$ ,  $x_2$ ) and a continuous target value (**Target**):

| Data Point | $x_1$ | $x_2$ | Target |
|------------|-------|-------|--------|
| A          | 1     | 2     | 3.5    |
| B          | 2     | 3     | 3.0    |
| C          | 3     | 1     | 4.5    |
| D          | 6     | 5     | 6.0    |
| E          | 7     | 8     | 7.5    |

### Step 1: Define the Test Point

Let's say we want to predict the target value for a new point **F** with coordinates  $x_1 = 4$  and  $x_2 = 4$ .

### Step 2: Choose the Value of K

Assume we choose  $K = 3$ .

### Step 3: Calculate the Distance

We'll calculate the Euclidean distance between point **F** and all points in the dataset.

### Euclidean Distance Formula:

$$d(p, q) = \sqrt{(X1_p - X1_q)^2 + (X2_p - X2_q)^2}$$

**Distance between F and A:**

$$d(F, A) = \sqrt{(4-1)^2 + (4-2)^2} \quad d(F, A) = \sqrt{(3)^2 + (2)^2} \quad d(F, A) = \sqrt{9+4} \quad d(F, A) = \sqrt{13} \approx 3.61$$

**Distance between F and B:**

$$d(F, B) = \sqrt{(4-2)^2 + (4-3)^2} \quad d(F, B) = \sqrt{(2)^2 + (1)^2} \quad d(F, B) = \sqrt{4+1} \quad d(F, B) = \sqrt{5} \approx 2.24$$

**Distance between F and C:**

$$d(F, C) = \sqrt{(4-3)^2 + (4-1)^2} \quad d(F, C) = \sqrt{(1)^2 + (3)^2} \quad d(F, C) = \sqrt{1+9} \quad d(F, C) = \sqrt{10} \approx 3.16$$

**Distance between F and D:**

$$d(F, D) = \sqrt{(4-6)^2 + (4-5)^2} \quad d(F, D) = \sqrt{(-2)^2 + (-1)^2} \quad d(F, D) = \sqrt{4+1} \\ d(F, D) = \sqrt{5} \approx 2.24$$

**Distance between F and E:**

$$d(F, E) = \sqrt{(4-7)^2 + (4-8)^2} \quad d(F, E) = \sqrt{(-3)^2 + (-4)^2} \quad d(F, E) = \sqrt{9+16} \\ d(F, E) = \sqrt{25} = 5.00$$

#### Step 4: Identify the Nearest Neighbors

Now, let's sort the distances:

| Data Point | Distance | Target |
|------------|----------|--------|
| B          | 2.24     | 3.0    |
| D          | 2.24     | 6.0    |
| C          | 3.16     | 4.5    |
| A          | 3.61     | 3.5    |
| E          | 5.00     | 7.5    |

The 3 nearest neighbors to point F are B, D, and C.

#### Step 5: Predict the Output

To predict the target value for F, we take the average of the target values of the 3 nearest neighbors:

**Target values of nearest neighbors:**

- B: 3.0
- D: 6.0
- C: 4.5

$$\text{Predicted Target for F: Predicted Target} = \frac{3.0+6.0+4.5}{3} = \frac{13.5}{3} = 4.5$$

Thus, the KNN algorithm predicts the target value for the point F as 4.5.